Leakage-Audited Multimodal Federated Learning for Tea Pest and Disease Diagnosis

Ayush Debnath, Ruelia Saha, Sudip Misra, Damodhara R. Mailapalli

***Abstract*—Tea pest and disease diagnosis draws on both lesion imagery and written field notes, but small agricultural corpora make multimodal and privacy claims easy to overstate. We report a leakage-audited comparison of five vision–language systems on one five-class tea corpus, under source-grouped splitting, exact image-box/text linkage, missing-modality tests, reliability-aware fusion, and measured FedAvg adaptation. The corpus holds 371 crops from 200 photographs, split 222/74/75 across 122/40/38 disjoint sources; every record links to its exact box, and a training-fitted filter masks 178 target-shortcut tokens. On the identical 75-crop support, frozen DistilBERT, frozen ViT-tiny, the raw ResNet–Transformer VLM, a frozen ViT-tiny–DistilBERT VLM, and the routed ResNet–Transformer reach 65.33%, 66.67%, 77.33%, 82.67%, and 81.33% accuracy. Selection is validation-only: the ViT–DistilBERT candidate's 82.43% validation accuracy stays below the routed incumbent's 89.19%, so the incumbent is retained. On test the candidate leads by a single crop out of 75, and its higher macro F1 (0.8192 versus 0.6679) traces almost entirely to one crop in a two-sample class. A same-checkpoint diagnostic falls from 92.00% to 33.33% once the note is class-mismatched, and on the locked test deleting the note moves 46 of 75 predictions while replacing the image with noise moves 27, so both branches are used and neither modality is advanced as decisive. The validation-selected ResNet–compact-Transformer VLM retains 98.1–100.0% of its 0.9099 centralized validation macro F1 under warm-start FedAvg over 2–8 simulated clients. Across five federated architectures on 371 genuine box-linked observations, federation outperforms local-only training, although the best aggregator is architecture-dependent; one float32 round costs 18.9–67.2 MiB per client depending on architecture. Two architectures saturate at 1.000 macro F1, which bounds what this corpus can establish; clients are simulated, so multi-site validation remains necessary.**

***Index Terms*—Federated Learning, Vision-Language Models, Tea Pest and Disease Classification, Multimodal Learning, Evaluation Protocol, Data Leakage, Precision Agriculture**

I. Introduction

TEA (*Camellia sinensis*) is a vital cash crop for millions of smallholder farmers across South Asia, East Africa, and East Asia. Five visually confusable stress classes — leaf blight, leaf hoppers, leaf rust, looper caterpillars, and mosquito bug — reduce yield and leaf quality across major growing regions [1], [2]. Recent diagnostic systems improve field deployment through lightweight CNNs, attention modules, and edge-oriented pipelines [1]–[3], [5], [6], [8], but nearly all begin and end with the photograph. That is not how tea diagnosis is recorded on the ground, where staff note lesion shape, flush condition, weather history, and pest pressure alongside the image. Meanwhile, sending every high-resolution image and field note to a central repository is unattractive for estates with limited connectivity and proprietary records. Federated Learning [15] addresses the second problem, but existing FL work on tea pathology has not integrated the visual and textual evidence used in real inspections.

A second, methodological motivation became dominant during this study. One source photograph may contain several annotated regions, so crop-level random splits place near-identical leaf context on both sides of the boundary. Text can leak the target even more directly when disease names or class-pure terms remain in descriptions. We therefore group by source image, link text at the exact box level, and learn the blocked vocabulary from training annotations only. This changes the scientific question from "can a model memorise a disease description?" to "does each modality contribute on unseen source leaves?"

Contributions. This is a comparison study, not a systems proposal: no architecture below is advanced as best, and the core carried into the routing and federated experiments is the one validation selected. We provide (i) exact crop–note linkage with source-disjoint splits and train-only leakage control; (ii) common-support unimodal, paired, mismatched, and model-family comparisons, extended by missing-modality and graded-corruption probes that test whether either branch is used at all; (iii) explicit visual and sparse-note routing; and (iv) three-seed client scaling plus a matched reviewer suite covering partitioning, anti-collapse components, warm start, fusion, communication, and four FL aggregators against local-only.

II. Related Work

Image recognition. Tea and plant disease work has moved from backbone benchmarking [3], [4] toward efficiency and deployment: attention-guided lightweight detectors [1], [45], [48], ensembles [6], [7], transformer hybrids [46], [49], real-time field datasets [5], IoT-assisted management [8], edge distillation and quantization [38], [40], and domain adaptation [39], [41]. Surveys map the breadth [2], [9]; across it the work is image-only, centralized, and split at crop or patch level, so one photograph can place near-identical regions on both sides of the boundary.

Agricultural federation. FedAvg [15] is extended with explainability [10], hierarchical sensing [11], blockchain-backed updates [12], cloud coordination [13], and coordinator-free edge federation [14], with trust and continual-update duties reviewed in [42], [47]. Those clients are almost always image-only, and federation is reported as a headline accuracy rather than a measured cost, so communication per round, warm start, realized skew, and matched-budget aggregator comparison are rarely reported together; we report them here.

Vision–language fusion. Contrastive alignment [22], query transformers over frozen encoders [23], gated cross-attention [24], joint captioning [25], and unified interfaces [26] supply the fusion vocabulary, realized here through compact encoders [17], [19], [29]–[33]. Agricultural instances include multimodal LLM fine-tuning [27], federated feature replay over frozen CLIP [28], sensor–image fusion [43], and knowledge-driven semantic classification [44]. Their recurring weakness is the text side: notes generated from the class label or paired by class matching are partial copies of the target. Ours is templated too, which is why pairing is fixed at the exact box index and the shortcut vocabulary is fitted on training text alone. Table I marks the gap: exactly paired image–text FL with a source-disjoint leakage audit.

TABLE I: Comparison with Related Work.

Work	Image	Text	FL	VLM	Leak. audit
Yang et al. [1], Hu et al. [2], Deng et al. [3]	✓	×	×	×	×
Joseph et al. [5], Maurya et al. [6]	✓	×	×	×	×
Tahir et al. [10], Devaraj et al. [11]	✓	×	✓	×	×
Ding et al. [12], Caminha et al. [13]	✓	×	✓	×	×
Thonglek and Rattanatamrong [14]	✓	×	✓	×	×
Kethineni et al. [44]	✓	✓	×	×	×
Wang et al. [27]	✓	✓	×	✓	×
Li et al. [28]	✓	✓	✓	✓	×
This work	✓	✓	✓	✓	✓

TABLE II: Source-grouped paired-data audit

Split	Sources	Crops	Blight	Hoppers	Rust	Looper/Mosq.
Train	122	222	77	5	40	61/39
Validation	40	74	26	2	14	20/12
Test	38	75	26	2	13	21/13
Total	200	371	129	9	67	102/64

III. System Model

Each estate keeps images and notes local; only model updates cross the boundary. The controlled ablations use one selected checkpoint, while FL adapts its trainable multimodal states with the visual backbone frozen. Text uses a stable 30 522-id hash vocabulary (length 128); normalized 224×224 images yield 49 spatial tokens for bidirectional attention.

A. Dataset and Leakage Audit

The audited corpus contains 200 field photographs collected from the IIT Kharagpur tea garden and annotated with YOLO oriented bounding boxes. One crop is extracted per valid box with 10% context padding, giving 371 examples across *LEAF_BLIGHT* (129), *LEAF_HOPPERS* (9), *LEAF_RUST* (67), *LOOPER_CATERPILLARS* (102), and *MOSQUITO_BUG* (64). Classes are the raw YOLO identifiers under the supplied annotation schema.

Seed 42 splits source filenames, never crops: overlap is zero and the test holds 75 crops from 38 unseen photographs. All 371 records link by exact (photograph, box index); the nine-example leaf-hopper class remains the bottleneck. Notes are templated from manually written phrase pools, not independent farmer prose. A 178-token target-pure vocabulary is learned on training text only and then frozen. No external tea images are pooled; ImageNet supplies weights only.

B. Model Variants and Objective

Table III summarizes the implementation. The reliability network produces (r_t, r_v), $r_t + r_v = 1$, and the final logits are $\mathbf{z} = \mathbf{z}_f + 2(r_t\mathbf{z}_t + r_v\mathbf{z}_v)$, with a missing modality's expert zeroed. The residual remains active at test time.

Comparison models. Frozen DistilBERT [33] and ViT-tiny [17] use validation-selected ridge heads. Their late-fusion candidate uses train-fitted 64-component projections and one of 360 validation-screened linear heads; test embeddings are extracted only after the choice is frozen. The raw and routed VLM rows are the cross-attention output before and after the router in Figs. 2 and 3.

For one-hot label $\mathbf{y}$, training uses class-weighted, label-smoothed cross entropy with auxiliary parent losses and supervised cross-modal alignment:

$$\mathcal{L} = \mathcal{L}_f + 0.20\mathcal{L}_t + 1.25\mathcal{L}_v + 0.05\mathcal{L}_{align}. \tag{1}$$

The visual weight is deliberately larger because image-only learning is the harder path. Modality dropout samples image-only, text-only, and paired batches with probabilities 0.50, 0.10, and 0.40. Checkpoint selection uses $0.65F_{1,\text{paired}} + 0.35F_{1,\text{image}}$ on validation.

TABLE III: Implemented model blueprint

Block	Specification	Output / training state
Text encoder	192-d embedding; 2 Transformer layers; 4 heads	256-d pooled vector; trainable
Vision encoder	ImageNet ResNet-50; 224^2 input	49×256 tokens; frozen backbone
Cross-attention	Two directional layers; 8 heads	text-to-image and image-to-text summaries
Reliability fusion	normalized (r_t, r_v) plus interaction features	5-way trainable classifier
Expert residual	linear text and vision heads; fixed $\lambda = 2$	added to the fusion logits at train and test
Complete model	32.20M parameters	8.69M trainable parameters

TABLE IV: Single visual models on the common support, ranked by the pre-declared validation score. *Cand.* is the number of validation-fitted configurations each system chooses among. Gates route per predicted class and are diagnostics, not deployable single models.

Visual system	Cand.	Val. acc.	Val. score	Test acc.	Test F1
Direct ResNet path	1	0.662	0.644	0.613	0.543
Six-view TTA	25	0.730	0.695	0.613	0.574
Deep-feature head	40	0.743	0.698	0.627	0.540
Frozen ViT-tiny	210	0.797	0.714	0.667	**0.583**
Calibrated base	20	0.770	**0.771**	**0.680**	0.570
Per-class gates (not single models)					
Three-expert gate	243	0.797	0.757	0.667	0.622
Four-expert gate	1024	0.811	0.783	0.693	0.653
Five-expert gate	3125	0.851	0.818	0.720	0.676

For K clients, the same local objective is aggregated with FedAvg [15],

$$\theta^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n}\theta_k^{t+1}, \tag{2}$$

where n_k is client k's sample count. FL starts from the selected checkpoint, freezes ResNet-50, and averages the remaining states after one local epoch; it is adaptation, not training from scratch.

IV. Evaluated Systems

A. Base Cross-Modal Checkpoint

The checkpoint trains for at most 15 AdamW epochs (batch 16, peak 10^{-4}). Two cached augmentations per training crop remain views of 222 crops, not new samples. Selection uses 74 validation crops; the strict-reloaded checkpoint is then evaluated once on 75 test crops.

B. Enhanced Image-Only Inference

Five single deployable visual systems compete on the common support. Table IV ranks them by the pre-declared validation score $0.6F_1 + 0.4$ accuracy, then reports the locked test support after the choice is frozen.

The calibrated base wins on both sides of the split — the validation score (0.771) and the highest single-model test accuracy (0.680, 51/75) — on the smallest search in the table. The gates score higher, but *Cand.* explains them: validation accuracy climbs monotonically with the routings fitted on 74 crops, 0.797 at 243 candidates to 0.851 at 3125, while the validation-to-test drop stays near 13 points throughout. The five-expert gate converts 3125 fitted choices into three test crops (54/75 against 51/75), inside the ±0.10 Wilson half-width here. We carry the calibrated base as the visual path and report the gate as a diagnostic ceiling.

C. Reliability-Aware Sparse-Note Rule

For sparse notes, a stable hash retains each non-special token with probability 0.50. A validation-fixed rule routes low-confidence (< 0.60), base-predicted leaf-rust cases to the

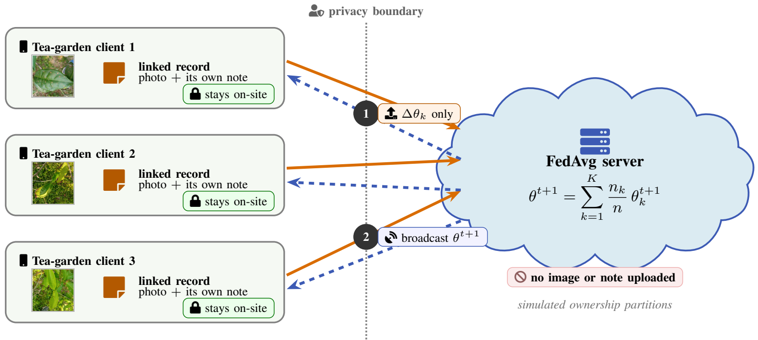


Fig. 1: Federated deployment workflow using representative corpus photographs. Each simulated tea-garden client keeps linked photographs and field notes on-site; only model updates $\Delta\theta_k$ cross the explicit privacy boundary, and the FedAvg server returns θ^{t+1}.

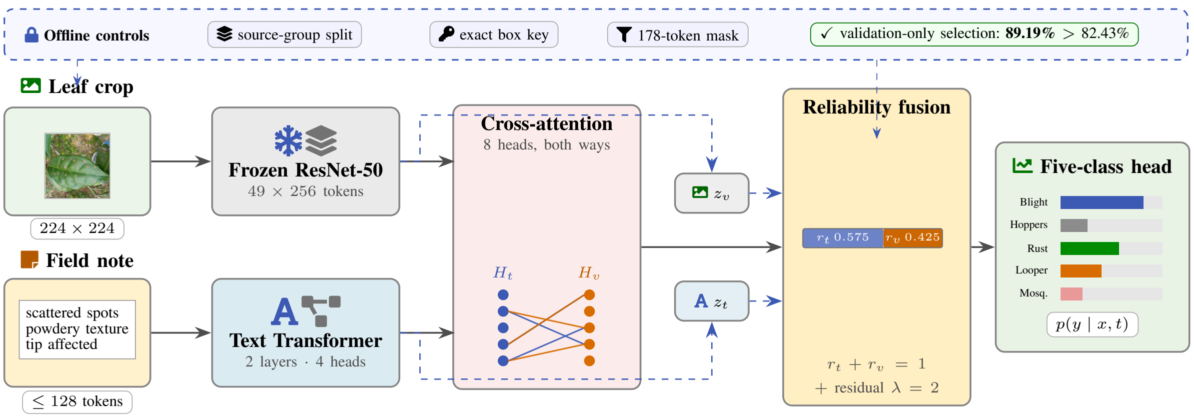


Fig. 2: Active inference architecture with a real corpus crop and a representative templated note. The dashed blue boxes separate offline leakage controls and validation-only system selection from runtime inference. The selected ResNet–compact-Transformer core uses bidirectional cross-attention; dashed runtime paths are explicit unimodal expert heads that re-enter the fusion logits. The pooled ViT–DistilBERT alternative is screened separately and is not drawn as token-level cross-attention.

sparse-text expert; otherwise it keeps late fusion. Because the routing family was refined on this internal partition, its test score is **exploratory and non-pristine**.

D. Multimodal Federated Adaptation

FL adapts 8.69M parameters with frozen ResNet-50, one local epoch, full participation, and sample weighting. We sweep $K = \{2, 3, 5, 8\}$ for eight rounds over seeds 42, 123, and 2026 on the 74-crop validation set. Clients are simulated; the ensemble and router are excluded from this FL score.

Figure 4 separates the two halves of a round: every stage touching a photograph, box, or note runs on the client.

E. Advisory Module

The Classify–Retrieve–Advise module embeds a query with Sentence-BERT [35] and performs exact FAISS inner-product search [34], [36]. Its separate templated corpus collapses 3,000 annotations to 673 observations; a deduplicated split leaves 472 passages and 201 queries with no shared observation. Correctness requires a class match.

V. Performance Evaluation

A. Protocol

The base checkpoint was trained in PyTorch [37] for at most 15 epochs. All crops from a source photograph remain in one partition. Model selection and the five-expert visual gate use only the 74-crop validation partition; every number below uses the same 75-crop, 38-source internal test support.

TABLE V: Validation-only multimodal FedAvg adaptation. Values are means over three client-partition seeds.

K	Final F1	Best F1	Retained	Label TV
2	0.8924	0.9099	98.1%	0.139
3	0.9046	0.9237	99.4%	0.217
5	0.9099	0.9297	100.0%	0.270
8	0.9099	0.9099	100.0%	0.300

Accuracy is accompanied by macro F1 because the class distribution is imbalanced. Source-group bootstrap intervals (5,000 replicates) resample source photographs rather than treating multiple crops from one photograph as independent observations.

Five evidence tracks remain separate: the exact-pair same-checkpoint ablation; the validation-selected family comparison on the common 75-crop support; the 74-crop validation-only FedAvg sweep; earlier text, vision, and proxy-fusion screens whose differing supports permit only within-panel ranking; and the 1,000-pair matched-setting reviewer suite. The last uses label-matched resamples and tests optimization and aggregation, not co-observation.

B. Federated Client and Round Scaling

Figure 5 and Table V answer the two federated scaling questions directly. Final-round macro F1 ranges from 0.8924 to 0.9099 and is never perfect. At the predeclared $K = 3$ setting, the final mean is 0.9046, or 99.4% of the 0.9099 initialization. Round-wise means stay between 0.8924 and 0.9216; the highest transient mean occurs for $K = 5$ at

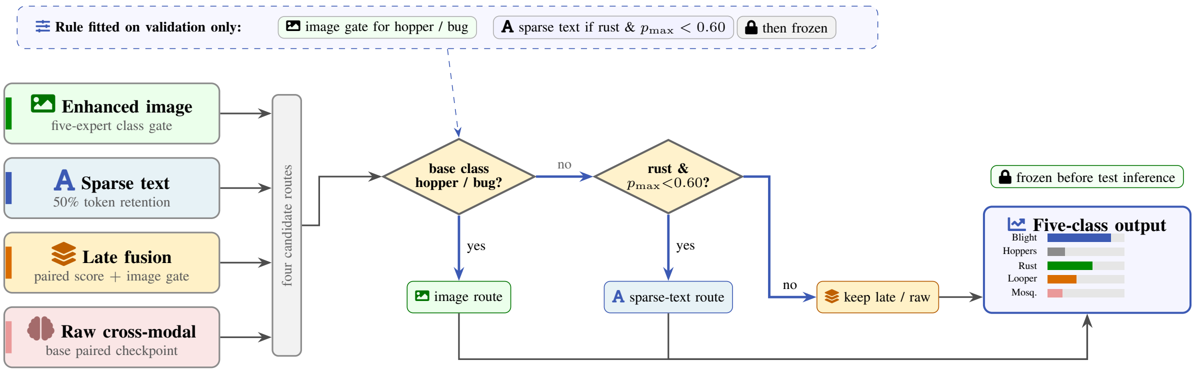


Fig. 3: Reliability-aware sparse-note routing as an explicit decision flow. Four available prediction routes feed a deterministic class/confidence rule selected on validation. Blue arrows show the frozen decisions; the router does not read test labels, and the reported internal result remains exploratory and non-pristine.

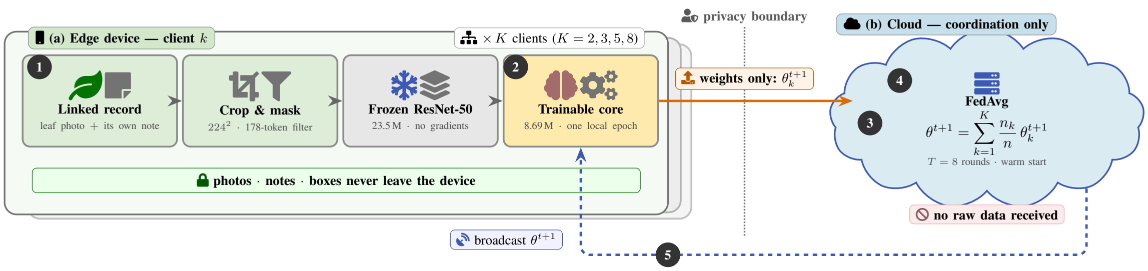
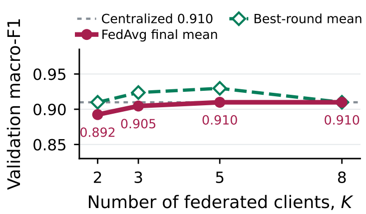
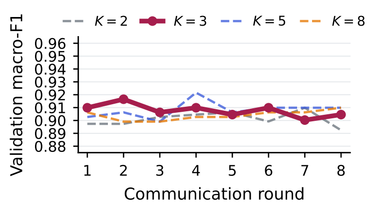


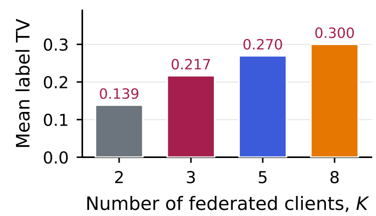
Fig. 4: Division of labour in one FedAvg round. The edge device runs every data-touching stage — box-linked loading, cropping and hashing, the shortcut filter, the frozen ResNet-50 pass, and one local epoch over 8.69M trainable parameters — while the server only receives, averages and broadcasts. No image, note, box, or per-sample quantity crosses the boundary; one float32 client-round costs 59.5–60.5 MiB up in the matched-setting suite.



(a) Client count versus final and best-round performance.



(b) Communication rounds; the deployed $K = 3$ trace is emphasized.



(c) Realized label skew actually induced at each client count.

Fig. 5: Measured multimodal FedAvg adaptation on the source-disjoint tea validation split. Points are mean macro F1 over three client-partition seeds; 222 training crops, 74 validation crops, $\alpha = 1$, one local epoch, full participation, and a frozen ResNet-50 are fixed. The dotted line in (a) is the 0.9099 initialization, and (c) is the realized label total variation, so (a) is read against measured rather than assumed skew. The 75-crop test is not used, and client ownership is simulated rather than estate-derived.

round 4, but it is reported as a validation peak rather than the final model. Increasing K raises realized label-distribution total variation from 0.139 to 0.300 without a monotone final-score penalty.

This stability result is useful but narrow. Every run begins from the same centrally selected checkpoint, so it supports federated adaptation of the multimodal core, not superiority over training from scratch. Three partition seeds measure ownership-split variability, not full checkpoint-training variance, and simulated clients cannot establish estate-level privacy.

C. Matched-Setting Reviewer Ablations

Figures 6 and 7, with Table VI, close the reviewer-requested gaps without mixing supports. The suite runs on 371 genuine box-linked observations, and within each cell every architecture trains on the identical client partition, so differences are architectural rather than partition effects. At $\alpha = 0.1$ the corrected splitter realizes label TV 0.451 against 0.265 for the legacy splitter, and federation beats local-only training for all five architectures (gains +0.121 to +0.348). Aggregator ranking is architecture-dependent: FedProx leads for the text, attention, and cross-attention models at $\alpha = 0.1$, while FedAvg leads for the image-only and concat models; at $\alpha = 1$ FedProx leads or ties throughout. SCAFFOLD is unstable in every configuration (0.088–0.579) and is always worst. Communication spans 3.5×, from 18.9 MiB per client-round for the image-only model to 67.2 MiB for cross-attention, so the image-only model reaches 0.499 macro F1 at roughly a third the uplink cost of either fusion model. The text-only and cross-attention models reach exactly 1.000 at $\alpha \geq 0.5$; on a 53-crop validation split this indicates that the templated notes are close to label-determining rather than that federation is

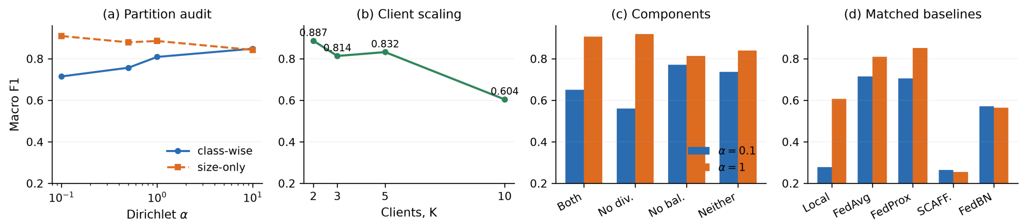


Fig. 6: Standard-tier reviewer suite. (a) The class-wise Dirichlet splitter creates the intended label skew; the legacy size-only splitter does not. (b) With fixed total data, $K = 10$ leaves much smaller client shards. (c) Anti-collapse components are ablated under actual federation. (d) Aggregators use the same backbone, partition, optimizer, and local budget. Bars in (c)–(d) compare $\alpha = 0.1$ and 1; E1/E8 average seeds 0 and 1, whereas E2/E3 use seed 0.

TABLE VI: Federated results on 371 genuine box-linked crop–note observations (260/53/58 over 153 independent source groups; no synthetic images). Every architecture within a cell trains on the identical client partition, so differences are architectural rather than partition effects. Values are two-seed means of validation macro F1.

(a) FedAvg by architecture and skew

Architecture	α=.1	α=.5	α=1	α=10
Image only	0.327	0.505	0.499	0.494
Text only	0.881	1.000	1.000	1.000
Concat VLM	0.640	0.760	0.761	0.778
Attention VLM	0.628	0.750	0.742	0.775
Cross-attn. VLM	0.816	1.000	1.000	1.000

(b) Aggregator at α=0.1

Arch.	Avg	Prox	SCAF	BN
Image	**.327**	.323	.211	.178
Text	.881	**.891**	.445	.753
Concat	**.640**	.638	.238	.570
Attention	.628	**.628**	.088	.588
Cross-att	.816	**.850**	.465	.773

(c) Cost and federation gain

Architecture	MiB	Δ local
Image only	18.9	+0.121
Text only	42.4	+0.348
Concat VLM	59.5	+0.312
Attention VLM	60.5	+0.288
Cross-attn. VLM	67.2	+0.339

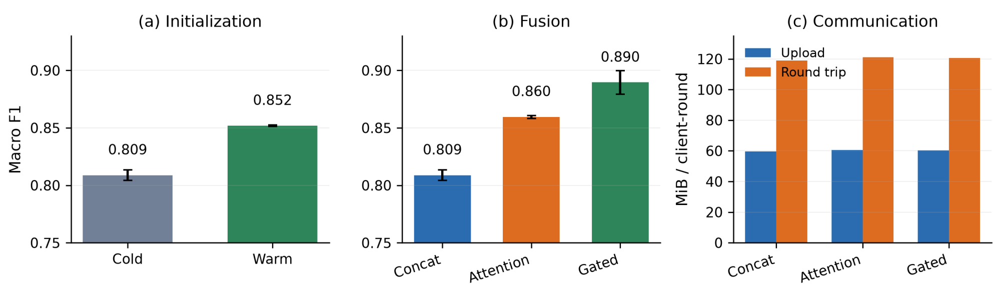


Fig. 7: Remaining matched-setting ablations (two-seed means with standard deviation bars): warm versus cold initialization, fusion strategy, and analytic float32 communication per client-round.

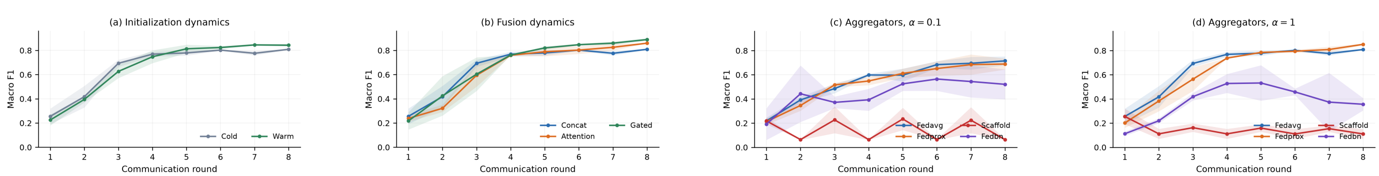


Fig. 8: Round-wise matched-setting dynamics. Lines are two-seed means and shading spans the observed seed range. Warm start separates late, gated fusion finishes highest, FedAvg/FedProx converge under both skews, and SCAFFOLD remains unstable on this support.

solved, and it bounds what the fusion comparison can establish on this corpus. Warm start adds 0.034 macro F1 at one seed and nothing at the other. The anti-collapse and fusion-variant ablations remain single-architecture and are reported for the concat model only.

Federated evidence previously covered one architecture. Table VII and Fig. 9 extend it to the frozen-encoder systems. Those systems use closed-form ridge heads centrally, which have no gradient updates for FedAvg to average, so they are federated here as linear probes: the encoder stays frozen and the head is trained by SGD. The federated head is therefore not the same estimator as the centralized ridge head, and the two are not directly comparable. Within this protocol fusion leads at both skews (0.697 and 0.831) and its federation gain over local-only training is the largest of the three systems in every cell (+0.187 to +0.446), so combining the branches helps more under partitioning than either branch alone. The frozen text encoder does not saturate here, unlike the scratch text model of Table VI, which supports reading that saturation as a property of the scratch encoder and the templated notes rather than of federation. Client partitions remain simulated.

TABLE VII: Federated adaptation across the paper's systems, not only ResNet+compact. Frozen encoders with SGD-trained heads on the shared source-grouped split; each system uses the identical client partition within a cell. Two-seed means of validation macro F1; local-only averages three clients.

	$\alpha = 0.1$		$\alpha = 1$	
System	FedAvg	Local	FedAvg	Local
Text (frozen DistilBERT)	0.579	0.337	0.628	0.527
Image (frozen ViT-tiny)	0.477	0.313	0.679	0.458
ViT+DistilBERT fusion	**0.697**	0.381	**0.831**	0.512
ResNet+compact (warm)	0.892–0.910 over 2–8 clients			

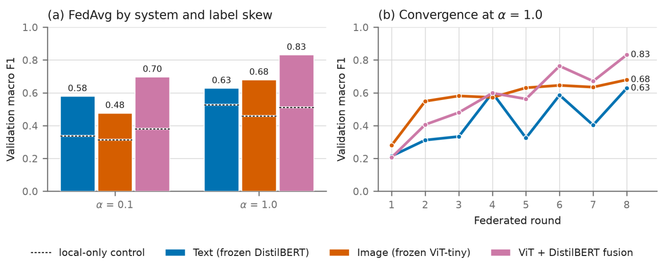


Fig. 9: Per-system federated adaptation. (a) FedAvg macro F1 by system and label skew; the dashed rule inside each bar is that system's local-only control, so the bar above it is the federation gain. (b) Per-round convergence at $\alpha = 1$. Every system in a cell trains on the identical client partition.

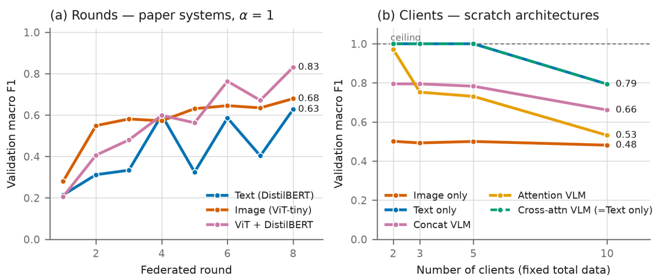


Fig. 10: Federated rounds and client scaling. (a) Per-round convergence of the paper's systems at $\alpha = 1$; fusion overtakes both parents after round five. (b) Macro F1 against the number of clients the same fixed training data is split across, for the scratch architectures. Text-only and cross-attention coincide at every client count, so the latter is drawn dashed; both sit at the 1.000 ceiling until $K = 10$, which is the saturation the text discusses rather than a scaling result.

TABLE VIII: Same-checkpoint ablation and configuration guidance. One core, one 75-crop support, two note conditions. Bold marks the configuration to use for the evidence each block assumes; the reversal between blocks is the result.

Condition	Accuracy	Correct	Macro F1	ECE
Complete notes — use text alone				
Core image path (ResNet)	0.6133	46/75	0.5431	0.1135
Core text path (compact Tr.)	**0.9600**	72/75	**0.9049**	0.1463
Base paired VLM	0.9200	69/75	0.8753	0.2648
Mismatched-pair VLM	0.3333	25/75	0.2804	0.2193
Half-masked notes — use the router				
Text alone	0.6400	48/75	0.5798	–
Visual gate	0.7200	54/75	0.6759	–
Plain fusion	0.7867	59/75	0.6472	–
Routed fusion	**0.8133**	61/75	0.6679	–

D. Standard Same-Checkpoint Ablation

Fusion is not uniformly worth its cost, and the two blocks say when it is. On complete notes it does not beat its strongest parent — 69/75 against 72/75 for text — so the text path alone is the configuration to deploy, and mismatching the note drops it from 92.00% to 33.33%, confirming cross-modal dependence. Masking half the note tokens reverses the ordering: text alone falls to 64.00% while the router holds 81.33%. The image earns its place only where the note is incomplete, which is the condition field notes are written under. With no note at all, the calibrated visual base of Table IV gives 68.00%. Two limits apply: notes are templated, so the complete-note block is the most inflated, and the routing family was refined on this partition, with three further hierarchy configurations placing the routed row between 73.33% and 81.33%.

TABLE IX: Common-support encoder selection. Architecture choice uses validation only; test metrics are reported after selection.

Actual system	Val. acc.	Test acc.	Test F1
Text PLM (DistilBERT)	0.6757	0.6533	0.6732
Image ViT (ViT-tiny)	0.7973	0.6667	0.5830
Raw ResNet+compact VLM	0.8378	0.7733	0.7155
ViT+DistilBERT VLM	0.8243	**0.8267**	**0.8192**
Routed ResNet+compact	**0.8919**	0.8133	0.6679

E. Raising Image-Only Accuracy

The validation-selected five-expert gate raises image accuracy from 61.33% (46/75) to **72.00%** (54/75; macro F1 0.6759). This is an ensemble, not the 66.67% pure-ViT baseline; rare-class behavior remains unstable.

F. Actual Model-Family and Inter-Model Comparison

The routed ResNet–compact model is the *incumbent* — the core already carried into the routing and federated experiments — and the pooled ViT–DistilBERT model is the *candidate*, adopted only if it beats the incumbent on validation. Validation selects the incumbent before test extraction: 89.19% accuracy and 0.8541 F1 versus 82.43% and 0.6892 for the candidate. The candidate later predicts one more test crop, but the 11-versus-10 discordance is not significant (McNemar $p = 1.0$). Figure 11 makes the selection/test distinction visible. Its top row contains earlier screens on different supports and therefore licenses within-panel ranking only.

The macro-F1 ordering reverses the accuracy ordering, and one two-crop class produces almost all of it. On leaf hoppers the incumbent scores F1 0.667 on validation and 0.000 on test, while the candidate scores 0.000 on validation and 0.667 on test — the two models swap a single crop, which accounts for 0.133 of the 0.151 test macro-F1 gap. Dropping that class, four-class macro F1 is 0.901 against 0.862 on validation and 0.835 against 0.857 on test, so the lead changes hands by 0.02–0.04. The incumbent's validation-to-test accuracy movement (66/74 to 61/75) is 1.35 standard errors. Macro-F1 differences on this support are therefore reported and not ranked.

G. Cross-Modal Analysis and Evidence Boundary

Mean reliability is 0.575 text/0.425 image; mismatching text costs 58.67 accuracy points and 0.5949 F1. Exact cross-modal recall@1 is only 1.33%, while class-level recall@1 is 0.600/0.613, indicating class organization rather than crop identity.

H. Missing-Modality and Corruption Probes

The mismatch result above shows dependence on the note; it does not show whether the image is used at all. We therefore evaluate the retained checkpoint on the locked 75-crop test under graded corruption of each modality and under two full image ablations, comparing predictions crop-by-crop against the clean run.

Both branches are load-bearing. Table X reports the sweep. Deleting every attended text token costs 0.551 macro F1 and moves 46 of 75 predictions; replacing the image with uniform noise costs 0.222 and moves 27. The reliability weight is not constant: mean w_t is 0.575 on clean input, 0.481 under complete text deletion, and 0.615 under complete image noise. The note carries more of the decision than the image, but neither branch is inert. No ordering of modalities or architectures is advanced from a single 75-crop support.

Ablation strength determines the verdict. Additive Gaussian noise at σ=0.35 leaves leaf and lesion plainly visible and

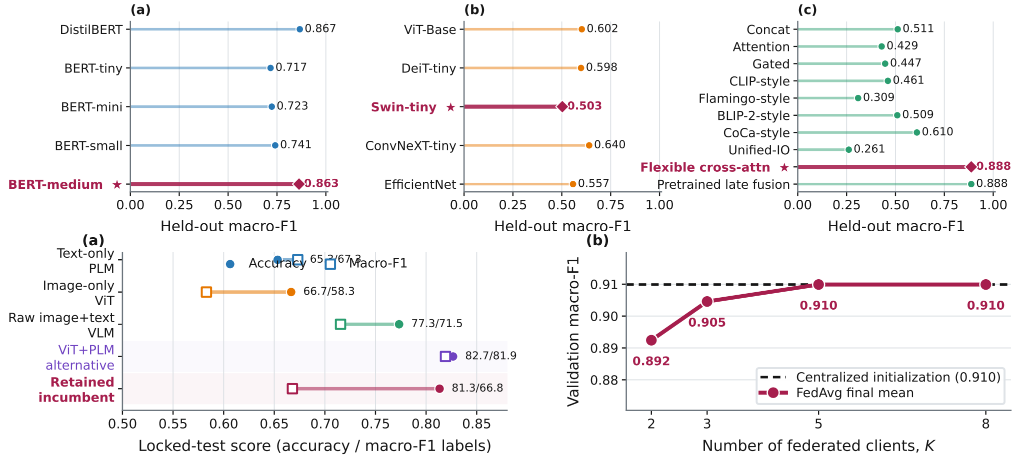


Fig. 11: Model comparisons retained as visual evidence. Top: within-family screens on their own supports (cross-panel scores are not comparable). Bottom: common 75-crop support for selected PLM, ViT, raw VLM, candidate VLM, and retained system (left), plus three-seed warm-start FedAvg scaling (right).

TABLE X: Modality corruption on the locked 75-crop test, retained checkpoint. Flips count predictions differing from the clean run.

Condition	Acc.	Macro F1	Flips
Clean	0.9200	0.8753	—
Text deletion 25%	0.8133	0.7445	10/75
Text deletion 50%	0.7333	0.6093	21/75
Text deletion 75%	0.6800	0.6275	24/75
Text deletion 100%	0.3600	0.3240	46/75
Image Gaussian σ=0.09	0.8533	0.7840	7/75
Image Gaussian σ=0.18	0.8933	0.8567	9/75
Image Gaussian σ=0.26	0.8667	0.8279	11/75
Image Gaussian σ=0.35	0.8933	0.8513	9/75
Image zeroed	0.8267	0.7544	18/75
Image uniform noise	0.6533	0.6535	27/75

moves only 9 of 75 predictions, whereas zeroing the image moves 18 and replacing it with uniform noise moves 27. An earlier 30-pair proxy screen used the Gaussian setting alone and read the visual branch as inert; that reading does not survive genuine removal on the locked support. Evidence that a branch is unused therefore requires an ablation that removes it rather than one that degrades it.

I. Architecture Component Ablation

Nine variants were retrained on the same split in CPU and H100 runs. Although the H100 full model reaches 94.67%, the identical CPU run gives 85.33%; the mean gap is four points and five of eight effects reverse sign. Figure 16c therefore reports instability, not a component ranking.

J. Advisory Retrieval Results

Retrieval alone returns a correct-class passage for 90.05% of queries (95% CI 85.57–94.03) against a 20.01% chance rate, with precision@5 93.23%, MRR 0.9469, and NDCG@5 0.9621. The classify stage raises this to 95.52% (95% CI 92.54–98.01), but that second number *is* the classifier's accuracy: retrieval restricted to one predicted class can only return that class, so routing rather than ranking decides correctness. Fig. 14(b) locates the failure. On the 65.67% of queries carrying a class-unique sentence open retrieval is near-saturated at 99.24%; on the rest it falls to 72.46%, which routing lifts to 89.86%. Per class it is weakest on looper caterpillars (77.50%), whose vocabulary overlaps mosquito bug damage.

Three limits apply. The corpus is template-composed, so base and queries share sentence vocabulary even though no observation is shared; these are upper bounds relative to free-form notes. The fusion re-ranker needs a fusion model on these five classes and is still unevaluated, as are the OBB overlay and the agronomic correctness of the advice itself. This is a retrieval result, not evidence that the recommended treatment is right.

K. Reproducibility Manifest

The updated bundle records data identity, checkpoint state, every federated round, and each realized client partition. Table XI separates mechanically verified controls from properties outside the evidence. Deterministic grouping, strict reload, and validation-only adaptation are reproducible; external estate transfer is not inferred from them.

TABLE XI: Reproducibility manifest for the updated run

Control	Recorded value	Status
Source split	seed 42; 122/40/38 sources	fixed
Cross-split overlap	0 source photographs	verified
Crop–text pairing	371/371 exact box keys	verified
Shortcut vocabulary	178 tokens; train fitted	fixed
Checkpoint reload	strict key match	passed
Architecture ablation	9 variants; CPU/H100 reruns	single seed each
DistilBERT selection	90 candidates; validation only	fixed before test embed.
ViT+PLM VLM screen	360 candidates; validation accuracy	incumbent retained
Visual gate search	3,125 gates on validation	completed
Internal test support	75 crops / 38 sources	common
FedAvg adaptation	$K = 2, 3, 5, 8$; 8 rounds	completed
FL repetitions	seeds 42/123/2026	partition seeds
FL evaluation	74 validation crops	test untouched
Reviewer suite	1,000 pairs; 700/150/150 split	label-matched proxy
Matched FL baselines	4 aggregators + local-only	completed; no errors

L. Class-Level Error Accounting

The enhanced gate gains eight correct crops and the exploratory router gains 15, mainly in blight and looper; two leaf-hopper examples remain too few for a stable claim.

M. Threats to Validity

The locked test is small (38 sources, 75 crops, two leaf-hopper crops), notes are templated, and selecting five experts on 74 validation crops can overfit. The central checkpoint has one seed; FL seeds vary only client partitions. Clients are simulated, and the reviewer suite uses class-matched resamples

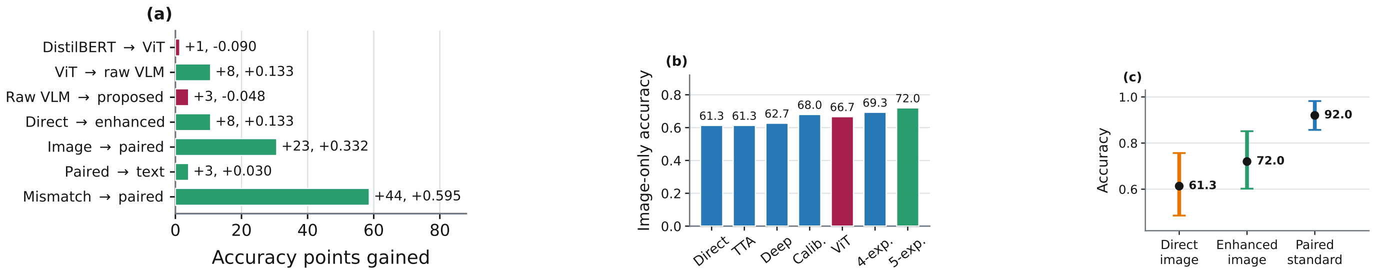
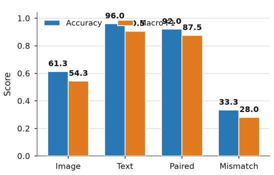
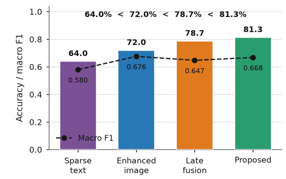


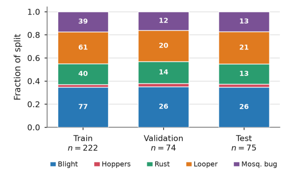
Fig. 12: Diagnostic evidence on the locked 75-crop support: (a) score gains, (b) image-expert progression, and (c) grouped bootstrap intervals. Visual gains are concentrated rather than uniform.



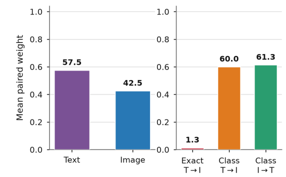
(a) Controlled paired-input ablation.



(b) Sparse-note routing hierarchy.



(c) Grouped split class balance.



(d) Reliability and retrieval probes.

Fig. 13: Compact diagnostic panels. (a) and (c) are controlled audited data; (b) is exploratory; (d) distinguishes class-level alignment from exact-pair retrieval. Metrics with different supports are not pooled.

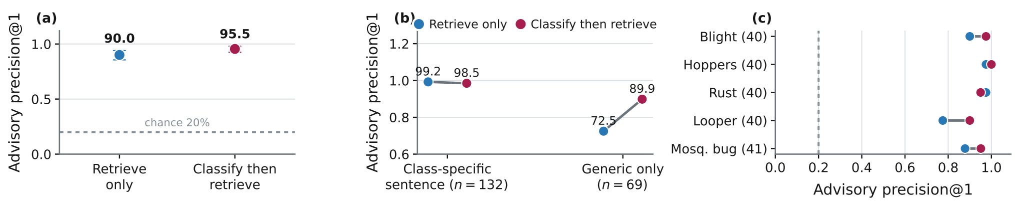


Fig. 14: Measured Classify–Retrieve–Advise quality on the templated symptom corpus: 472 passages, 201 queries, exact inner-product search. (a) Precision@1 with bootstrap intervals against chance; (b) split by whether the query holds a class-unique sentence; (c) per class. Same five classes as the diagnostic experiments, but a different corpus and support, never pooled with them.

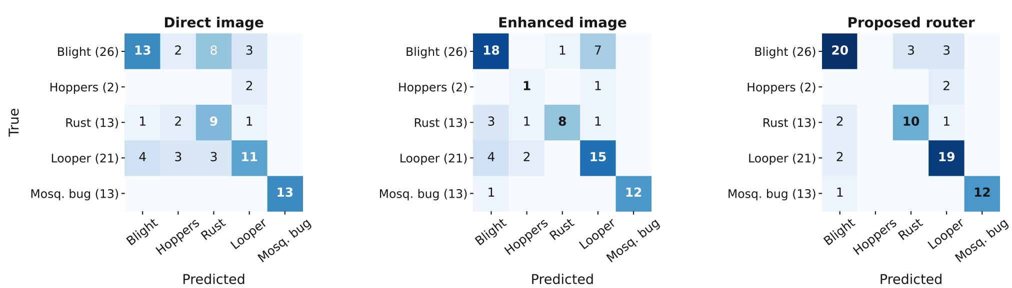
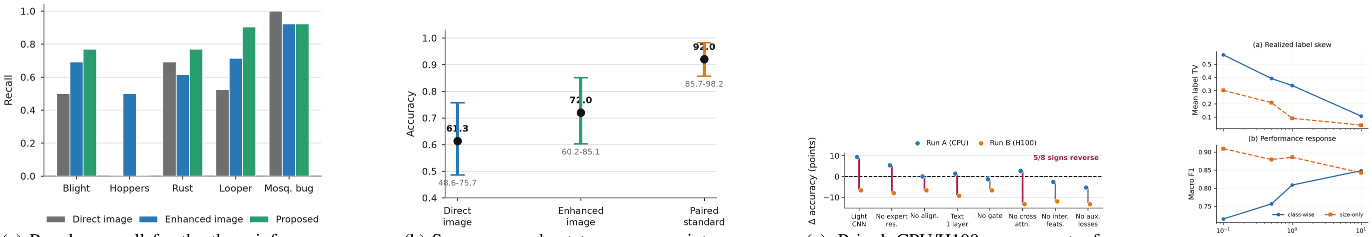


Fig. 15: Where the errors land on the fixed 75-crop test. Each matrix sums to 75; diagonals give the 46, 54, and 61 correct crops of the direct image path, the five-expert enhanced gate, and the exploratory router. All panels share one count colour scale, so shading is directly comparable. The off-diagonal cells are the part no count table records: leaf blight is confused with rust and looper rather than scattered, looper recovers most under the router, and the two leaf-hopper crops fall into looper except for one the enhanced gate recovers.

with synthetic image fill rather than co-observations; E2/E3 use one seed. The router result is non-pristine, DistilBERT is a PLM rather than a generative LLM, and earlier fusion screens on proxy pairs have incomparable supports, so their patterns are read without their levels. The corruption probe is one seeded realization per severity on 75 crops and carries no interval. Deployment requires multi-estate data, independent notes, rare-class coverage, and repeated end-to-end training.

VI. Conclusion

On 371 linked crops, validation retains the ResNet–compact-Transformer router; the ViT–DistilBERT candidate's one-crop test edge is non-significant ($p = 1.0$), and CPU/H100 component effects are unstable. Warm-start FedAvg retains 98.1–100.0% of initialization over 2–8 simulated clients. In the matched suite, FedAvg reaches 0.715/0.809 at $\alpha = 0.1/1$, warm start adds 0.043, and updates cost about 60 MiB. No E3 arm collapses, so the stack is not claimed necessary. Multi-estate, co-observed evaluation remains required.



(a) Per-class recall for the three inference paths.

(b) Source-group bootstrap accuracy intervals.

(c) Paired CPU/H100 component effects.

(d) Partition audit.

Fig. 16: Class-level gains, uncertainty, component stability, and partition audit on the locked 75-crop test. Intervals in (b) resample 38 source photographs and do not measure estate shift. In (c) red connectors cross zero and five of eight signs reverse, so single-run deltas are not ranked. In (d) the legacy size-only splitter stays comparatively IID.

References

[1] J. Yang, G. Xu, M. Yang, and Z. Lin, "Lightweight wavelet-CNN tea leaf disease detection," *PLOS ONE*, vol. 20, no. 5, art. e0323322, 2025.

[2] Y. Hu, J. Tang, and J. Yang, "Introducing artificial intelligence technology to plant disease management for sustainable agriculture," *Crop Protection*, vol. 184, art. 106764, 2024.

[3] X. Deng and C. Photong, "Evaluation of tea leaf disease identification based on CNNs VGG16, ResNet50, and DenseNet169," in *Proc. iEECON*, 2024, pp. 1–4.

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.

[5] D. S. Joseph, P. M. Pawar, and K. Chakradeo, "Real-time plant disease dataset development and detection of plant disease using deep learning," *IEEE Access*, vol. 12, pp. 16310–16333, 2024.

[6] R. Maurya, S. Mahapatra, and L. Rajput, "A lightweight meta-ensemble approach for plant disease detection suitable for IoT-based environments," *IEEE Access*, vol. 12, pp. 28096–28108, 2024.

[7] Md. S. H. Shovon *et al.*, "PlantDet: A robust multi-model ensemble method based on deep learning for plant disease detection," *IEEE Access*, vol. 11, pp. 34846–34859, 2023.

[8] E. A. Al-Shahari *et al.*, "Internet of Things assisted plant disease detection and crop management using deep learning for sustainable agriculture," *IEEE Access*, vol. 13, pp. 3512–3520, 2025.

[9] W. Shafik, A. Tufail, A. Namoun, M. S. Abu Bakar, and R. A. A. H. M. Apong, "A systematic literature review on plant disease detection," *IEEE Access*, vol. 11, pp. 59174–59203, 2023.

[10] H. A. Tahir, W. Alayed, and W. U. Hassan, "A federated explainable AI framework for smart agriculture," *IEEE Access*, vol. 13, pp. 97567–97584, 2025.

[11] H. Devaraj *et al.*, "RuralAI in tomato farming: Integrated sensor system, distributed computing, and hierarchical federated learning," *IEEE Sensors Lett.*, vol. 8, no. 5, pp. 1–4, 2024.

[12] Q. Ding, X. Yue, Q. Zhang, Z. Xiong, J. Chang, and H. Zheng, "Bc2FL: Double-layer blockchain-driven federated learning for Agricultural IoT," *IEEE Internet Things J.*, vol. 12, no. 4, pp. 4362–4374, 2025.

[13] P. V. Caminha and H. M. N. da S. Oliveira, "Cloud-driven federated learning for plant disease detection in agriculture," in *Proc. IEEE CloudNet*, 2024, pp. 1–4.

[14] K. Thonglek and P. Rattanatamrong, "Decentralized federated learning for agricultural plant diseases identification on edge devices," in *Proc. iSAI-NLP*, 2023, pp. 1–6.

[15] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. AISTATS*, 2017, pp. 1273–1282.

[16] A. Vaswani *et al.*, "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–6008.

[17] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.

[18] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL*, 2019, pp. 4171–4186.

[19] I. Turc, M.-W. Chang, K. Lee, and K. Toutanova, "Well-read students learn better: On the importance of pre-training compact models," *arXiv:1908.08962*, 2019.

[20] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. CVPR*, 2016, pp. 2818–2826.

[21] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE TPAMI*, vol. 42, no. 2, pp. 318–327, 2020.

[22] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.

[23] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. ICML*, 2023.

[24] J.-B. Alayrac *et al.*, "Flamingo: A visual language model for few-shot learning," in *Proc. NeurIPS*, 2022.

[25] J. Yu, Z. Wang, V. Vasudevan, L. Yeung, M. Seyedhosseini, and Y. Wu, "CoCa: Contrastive captioners are image-text foundation models," *TMLR*, 2022.

[26] J. Lu, C. Clark, R. Zellers, R. Mottaghi, and A. Kembhavi, "Unified-IO: A unified model for vision, language, and multi-modal tasks," in *Proc. ICLR*, 2023.

[27] Y. Wang *et al.*, "A large language model for multimodal identification of crop diseases and pests," *Scientific Reports*, vol. 15, art. 21959, 2025.

[28] L. Li, J. Li, D. Chen, L. Pu, H. Yao, and Y. Huang, "FedReplay: A feature replay assisted federated transfer learning framework for smart agriculture," *arXiv:2511.00269*, 2025.

[29] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. ICML*, 2021, pp. 10347–10357.

[30] Z. Liu *et al.*, "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. ICCV*, 2021, pp. 10012–10022.

[31] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. CVPR*, 2022, pp. 11976–11986.

[32] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.

[33] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT," *arXiv:1910.01108*, 2019.

[34] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

[35] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP*, 2019, pp. 3982–3992.

[36] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. NeurIPS*, 2020, pp. 9459–9474.

[37] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. NeurIPS*, 2019, pp. 8026–8037.

[38] C. V. Padeiro, T.-W. Chen, T. Komamizu, and I. Ide, "SPQ: Similarity-preserving posttraining quantization for portable crop disease detection," *IEEE Trans. AgriFood Electron.*, vol. 4, no. 1, pp. 68–80, 2026, doi: 10.1109/TAFE.2025.3533416.

[39] S. A. A. Qadri, N. F. Huang, Y.-H. Huang, and P.-C. Chan, "DDAN-RGAL: A dual-domain adaptive network with reality-gap aware learning for field-deployable plant disease detection," *IEEE Trans. AgriFood Electron.*, early access, pp. 1–18, 2026, doi: 10.1109/TAFE.2026.3667568.

[40] Q. Tong, N. Zeng, S. Wu, and K. Xu, "Tiny-PMDet: A knowledge-distilled lightweight edge-AI detector for grapevine powdery mildew detection on CMOS sensor based AIoT systems," *IEEE Trans. AgriFood Electron.*, vol. 4, no. 1, pp. 383–394, 2026, doi: 10.1109/TAFE.2026.3667753.

[41] N. Nadeem, M. H. Asad, and A. Bais, "CCUDA-SS: Cross-crop unsupervised domain adaptation for semantic segmentation," *IEEE Trans. AgriFood Electron.*, early access, pp. 1–17, 2026, doi: 10.1109/TAFE.2025.3650340.

[42] G. M. Dimitri, A. Kocian, F. Scarselli, M. Gori, and S. Chessa, "Continual learning for agrifood: A systematic review," *IEEE Trans. AgriFood Electron.*, vol. 4, no. 1, pp. 117–127, 2026, doi: 10.1109/TAFE.2025.3616733.

[43] P. Garg, A. Mishra, R. Raja, A. Kumar, M. V. Joshi, and V. S. Palaparthy, "Multimodal data fusion by integrating IoT-enabled sensors and images for Jamun crop disease detection with machine learning," *IEEE Trans. AgriFood Electron.*, vol. 3, no. 2, pp. 582–590, 2025, doi: 10.1109/TAFE.2025.3585065.

[44] K. K. Kethineni, S. P. Mohanty, and E. Kougianos, "Semantic-Search: A knowledge-driven classification method for accurate identification of plant diseases," *IEEE Trans. AgriFood Electron.*, early access, pp. 1–11, 2025, doi: 10.1109/TAFE.2025.3619095.

[45] K. Cao, Y. Yang, L. Xu, X. Wan, L. Yang, and W. Jia, "GAF-Net: An enhanced multidisease for corn leaves detection model based on YOLOv8," *IEEE Trans. AgriFood Electron.*, vol. 3, no. 2, pp. 420–432, 2025, doi: 10.1109/TAFE.2025.3561196.

[46] S. Das, P. Bikram, A. Biswas, V. C., P. Sinha, and B. B. Bhattacharya, "Transformer-embedded attentive CNN for spectral image analysis of rice blast syndromes," *IEEE Trans. AgriFood Electron.*, vol. 3, no. 2, pp. 615–622, 2025, doi: 10.1109/TAFE.2025.3601808.

[47] M. A. R. Baee, L. Kane, L. Simpson, Y. Jiang, W. Armstrong, and P. Gauravaram, "Trustworthy digital agriculture: Challenges, countermeasures, and opportunities," *IEEE Trans. AgriFood Electron.*, vol. 3, no. 2, pp. 518–535, 2025, doi: 10.1109/TAFE.2025.3576623.

[48] B. Ji *et al.*, "Lightweight tomato leaf intelligent disease detection model based on adaptive kernel convolution and feature fusion," *IEEE Trans. AgriFood Electron.*, vol. 2, no. 2, pp. 563–575, 2024.

[49] P. S. Thakur *et al.*, "Real-time plant disease identification: Fusion of vision transformer and conditional convolutional network with C3GAN-based data augmentation," *IEEE Trans. AgriFood Electron.*, vol. 2, no. 2, pp. 576–586, 2024.